

VR-based Context Priming to Increase Student Engagement and Academic Performance

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ABSTRACT

Research suggests that virtual environments can be designed to increase engagement and performance with many cognitive tasks. This paper compares the efficacy of specifically designed 3D environments intended to prime these effects within Virtual Reality (VR). A 27-minute seminar “The Creative Process of Making an Animated Movie” was presented to 51 participants within three VR learning spaces: two prime and one no-prime. The prime conditions included two situated learning environments; an animation studio and a theatre with animation artifacts vs. the no-prime: theatre without artifacts. Increased academic performance was observed in both prime conditions. A UX survey was also completed.

Index Terms: Applied computing – Education – Computer-assisted instruction; Human-centered computing – Virtual reality

1 INTRODUCTION

Winston Churchill famously said, “We shape our buildings and afterwards our buildings shape us” [3]. We know instinctively that where we live, work, and learn can affect how we feel and how we perform. Nature environments and creative spaces with gorgeous aesthetics are highly coveted and can improve performance [10] yet are inaccessible for most people.

Virtual Reality (VR) allows designers to create and design these coveted spaces to prime (act as stimuli to improve cognition) academic performance. While there is research demonstrating that the context of our experiences can improve cognition [1][5], there is little research validating the transferability of these effects to more advanced educational technology like VR or research that informs the design choices that improve cognition within VR environments. Given the potential for VR priming to greatly impact the learning process and building on previous work on VR priming interventions [6], we investigate environment context priming: the impact of VR environments on learning.

Our results clearly demonstrate the positive effects of both prime scenarios to improve academic performance compared to the no prime condition. Following a review of related work and description of our study, we discuss the performance and UX results that will inform the design of future VR experiences.

2 RELATED WORK

2.1 Priming

Priming occurs when a stimulus (the prime) makes the content and subsequent cognitive processes more accessible [8]. Most often, priming stimuli are supraliminal (observable but not obvious) where changing the context of an environment can create effects with little or no perceptual awareness. A picture can create

bias and affect how a person thinks [7] or placing codes (A vs. F) on pre-test forms can affect academic performance [4].

2.2 VR as a “Prime” Enabler

Central to the concept of VR is the quantifiable concept of immersion and the more subjective idea of “presence” or the feeling of being there. Based on these immersive possibilities and the ability to create a sense of presence, VR is an ideal environment to create scenarios to evoke different forms of conscious awareness [11]. For designers, VR offers the possibility to create learning spaces that are more conducive to learning while “designing out” potential anxiety-inducing barriers. In a 2013 study, 3D virtual environments were used to test the effects of priming on creativity. The results showed that when teams created ideas in the primed virtual environments, they created more ideas and of higher quality than the control groups [1].

2.3 Learning Theory

Situated Learning Theory (SLT) [9] recognizes the value of social and contextual experiences within a community of practice. Within a specific situation learners can see, hear, do, and feel the experience resulting in higher retention and improved performance. Based on SLT, technology solutions that offer improved situational contexts and modalities should increase engagement and performance. For example, placing someone in the context of an animation studio could amplify the understanding of an animation related idea or concept [9][2].

3 METHODOLOGY

We conducted a user study to evaluate context priming in VR using a remote research process. VR headsets were provided to students in their own homes. The study was approved by the University’s Research Ethics Board and followed all guidelines for safe experimentation during the COVID-19 pandemic.

3.1 Study Overview

The study, a between-subjects test with one independent variable (Priming context), explored the effectiveness of context priming. A 27-minute seminar “The Creative Process of Making an Animated Movie” was presented to 51 participants within three VR learning spaces: two prime and one no-prime. The prime conditions included two situated learning environments; an animation studio and a theatre with animation artifacts vs. the no-prime: theatre without animation artifacts. The priming conditions were selected based on SLT [2][9], that emphasizes the role of authentic and realistic environments in learning. The study used two dependant variables (academic score and user experience). Academic score is our main subject and represents the learning effect of the context. It was measured through a short test after the course. Based on SLT, and the anticipated priming effects of the VR environments, we investigated the following hypothesis:

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H1: (a) Prime conditions (both) will improve academic performance over the no prime condition. (b) There will be no significant difference between two priming conditions.

H2: (a) Prime conditions (both) will improve assessments of UX results compared to the no prime scenario. (b) There will be no significant difference between two priming conditions.

3.2 Participants

The study was performed with 51 participants over three priming conditions (17 in each condition) between 18 and 73 with an equal balance of gender. They were randomly divided into three groups corresponding to prime or no-prime scenarios. Participants were paid \$20 CDN (via e-transfer) upon completion of the course and the post seminar questionnaires.

3.3 Apparatus

Participants were provided with an Oculus Quest or Quest 2 VR Head-Mounted Display (HMD) that auto-loaded the necessary testing software from the Oculus App Lab. Testing and UX questionnaires were completed on personal computers or mobile tablets within a standard browser. The 27-minute seminar "The Creative Process of Making an Animated Movie" was presented in an audio-visual format on a large screen with the professor presenting from behind the podium. The multiple-choice questions (20) related directly to the content presented in the seminar. The UX survey consisted of 28 questions related to the qualitative aspects of the experience and feelings of presence.

3.4 Procedure

Given the limitations caused by COVID-19, we designed a study that could be performed independently by participants at their homes. Before testing, pre-test consent forms were completed at www.toonrush.com/vrtest. The VR learning seminar was provided as a software app (APK) that was loaded automatically onto the VR headset using the Oculus App Lab developer utility. Upon completion of the seminar, participants returned to the website to complete the post-seminar UX survey and multiple-choice test related to the seminar content.

4 RESULTS

We had a 1x3 experiment design with one independent variable (Priming condition – prime vs. no prime) and two dependant variables (academic performance, UX satisfaction). After reviewing the outlier data, we opted to eliminate the highest value from each of the three conditions to avoid skewing results based on potentially contaminated data. These high values occurred for academic score and were more than twice the average. They came from participants who were very familiar with the subject. Hence our sample size was reduced to 16 participant inputs per condition. With an equal number of participants (16) we first conducted a single factor ANOVA test for academic score and data samples. After confirming significance of mean departure within the conditions ($p=.0099$), we performed a post-hoc t-test with 2 samples, assuming equal variance. We observed significance in academic performance for both prime conditions compared to the no prime condition (prime (artifacts) vs. no prime ($P=.0024$) and prime (animation studio) vs. no prime ($P=.0104$). There was no significance observed between the two prime conditions related to academic performance. While UX feedback on qualitative aspects of the experience was positive we did not observe significance in either case comparing the primes to the control condition.

5 DISCUSSION

In reviewing the first hypothesis, both prime conditions observed significant improvement over the no-prime condition, while there was no significant difference between two priming contexts. Hence, H1a and H1b are both supported. In reviewing the second hypothesis (UX), neither prime condition demonstrated significance over the no prime condition. Hence, H2a was not supported. The prime conditions observed no significance between themselves with respect to UX results. Hence, H2b is supported.

t-Test: Two-Sample Assuming Equal Variances	Artifacts Prime	No Prime	Animation Studio	No Prime
Mean	11.94	9.6875	11.44	9.6875
Variance	4.86	3.83	4.4	3.83
Observations	16	16	16	16
Pooled Variance	4.35		4.11	
Hypothesized Mean Difference	0		0	
df	30		30	
t Stat	3.05		2.44	
P(T<=t) one-tail	0.0024		0.0104	
t Critical one-tail	1.6973		1.6973	
P(T<=t) two-tail	0.0047		0.0208	
t Critical two-tail	2.0422		2.0423	

Figure 1: Priming Study Results.

6 CONCLUSION

Motivated by the disruptive potential of immersive VR, we wanted to understand the priming potential of VR environments to improve student performance. In our study of 51 participants, with three conditions (2 prime, 1 no prime) we observed a clear priming effect that we attribute to the situated learning effect of a simulated animation studio and subject matter artifacts situated within a VR learning space.

Thanks to Edin Ibric and Jon Jackson for design support services.

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